SEPM CCA PROJECT

Requirements & Analysis

College Event Management System

Person 1 Deliverable | CCA-Level Requirements & Analysis Document

Course	SEPM – CCA	Team size	4 members
Project tool	Jira	**Assigned role**	Person 1 – Requirements & Analysis

1. Project Overview

The College Event Management System is a proposed web-based application for organizing college events through one central platform. It is intended to support the main activities involved in an event: publishing event information, accepting student registrations, maintaining participant details, managing schedules and venues, and providing participation certificates to eligible students. The aim is to make event-related information easier to access and manage for both students and organizers.

1.1 Objective

The objective is to provide students with one place to discover college events, understand the event details and register for activities. For organizers, the system should reduce the effort involved in maintaining event information, checking participant lists, updating schedules and venues, and issuing certificates. The first version will focus on these core activities rather than advanced features.

1.2 Problem Being Addressed

Event information and registration details may otherwise be shared through separate notices, messages, online forms or manual lists. This can make it difficult for students to find the latest information and for organizers to keep participant details organized. A centralized system can bring the main event information and activities together, making the process clearer and easier to follow.

1.3 Expected System Flow

The organizer creates or publishes an event with its description, date, time and venue. Students browse the available events and register for an event that is open for registration. The organizer then views the registered participants and manages any schedule or venue updates. After the event, the organizer can issue certificates to eligible participants, who can access their certificates through the system.

2. Project Scope

The initial version will cover the core activities required to announce, organize and participate in college events. The following functions form the proposed project scope:

- Display college events and their basic details so students can see which events are available.
- Allow students to register for available events and receive a clear registration confirmation or message.
- Allow organizers to create, update and cancel event details so that the information shown to students remains current.
- Allow organizers to view and manage participant registrations for each event.
- Maintain event schedules and venue information so participants know when and where an event will take place.

- Allow authorized organizers to prepare or issue participation certificates for students who meet the event eligibility conditions.
- Allow eligible students to view or download their participation certificates.

The initial CCA version will not include online payments, QR-based attendance, AI recommendations, dedicated mobile applications or external SMS integrations. These features are excluded to keep the prototype manageable and focused on the main functions stated in the assigned problem statement.

3. Stakeholders / User Roles

The system has two main user groups: students who participate in events and organizers who manage them. An administrator role may also be included if the team decides that separate user and access management is needed. Each role is expected to access only the functions relevant to its responsibilities.

User role	Purpose	Main activities
Student / Participant	Find and participate in college events by viewing event information, registering and checking participation details.	Browse events and their details; register for an event; check registration status; view or download an eligible certificate.
Event Organizer / Coordinator	Plan and coordinate events and keep event information and participant details up to date.	Create or update events; view participant lists; manage schedules and venues; prepare certificates for eligible participants.
Administrator	Oversee basic user access and administrative functions, if this role is included in the prototype.	Manage user accounts or roles where required. This role is optional and can be kept limited for the CCA version.

4. Basic Requirement Elicitation Approach

Requirement elicitation means identifying what the people who use the system need it to do. For this CCA project, the process is kept simple and is based mainly on the assigned PS 2 problem statement and the basic workflows of students and event organizers. A formal survey or extensive interview study is not necessary for the current project scope, and this document does not claim that such a study was conducted. Any assumptions can be checked briefly with the team or faculty before the requirements are finalized.

4.1 Lightweight Elicitation Steps

The initial requirements are organized using the following simple steps:

Step	Activity	Purpose / Output
1. Review the problem statement	Read the assigned PS 2 description and identify the stated functions: registration, participants, schedules, venues and certificates.	Creates the initial feature list for the system.
2. Consider user roles	Look at the expected activities of students, event organizers and, if needed, an administrator.	Identifies who will use each feature and why.
3. Clarify basic workflows	Discuss any unclear steps with the four-member team, such as how a student registers or how an organizer updates an event.	Helps agree on a simple and practical first version.
4. Record and review requirements	Convert the identified needs into functional requirements, non-functional requirements, use cases and user stories.	Produces the initial requirements set for review and handover.

4.2 Main Needs Identified

From the problem statement and the proposed user roles, students need to find events, view key details, register and access certificates. Organizers need to publish and update event information, view participant registrations, maintain schedules and venues, and issue certificates. These needs form the basis of the functional requirements and use cases in the following sections.

4.3 Elicitation Output and Review

The output of this lightweight process is the initial list of functional and non-functional requirements, the main actors and use cases, and the user stories with acceptance criteria. The team should review these items together and remove or revise anything that is outside the agreed CCA scope. This review is the practical validation step before the requirements are shared with the design and development members.

5. Functional Requirements

Functional requirements describe the actions and services the system must provide. Each item below is written as a proposed system behaviour and is given a unique ID so that the team can refer to it during design, development and basic testing. The notes explain the purpose of each requirement in simple terms.

ID	Requirement	Note
FR-01	The system shall allow registered users to log in.	Provides a basic way to identify users and restrict access to relevant functions. Include this only if login is part of the agreed prototype.
FR-02	The system shall display the list of available college events.	Gives students a central place to discover events that are available for participation.
FR-03	The system shall display event details such as description, date, time and venue.	Shows the information a student needs before deciding whether to register, including the event date, time and venue.
FR-04	The system shall allow students to register for an available event.	Records a student's request to participate in a selected event while registration is open.
FR-05	The system shall prevent the same student from registering more than once for the same event.	Avoids repeated entries for the same student and event, keeping the participant list more accurate.
FR-06	The system shall allow organizers to create, update and cancel events.	Allows organizers to maintain correct event information and communicate changes through the system.
FR-07	The system shall allow organizers to view the list of registered participants for an event.	Helps organizers check who has registered for a particular event and plan participation.
FR-08	The system shall allow organizers to add or update event schedules.	Allows organizers to maintain event timing information and update it when required.
FR-09	The system shall allow organizers to assign or update an event venue.	Allows organizers to record the location where the event will be conducted and update it if needed.

ID	Requirement	Note
FR-10	The system shall allow authorized organizers to generate or issue participation certificates to eligible participants.	Allows an authorized organizer to prepare certificates after confirming which participants are eligible.
FR-11	The system shall allow eligible participants to view or download their certificates.	Provides eligible participants with access to their participation certificate after it is issued.

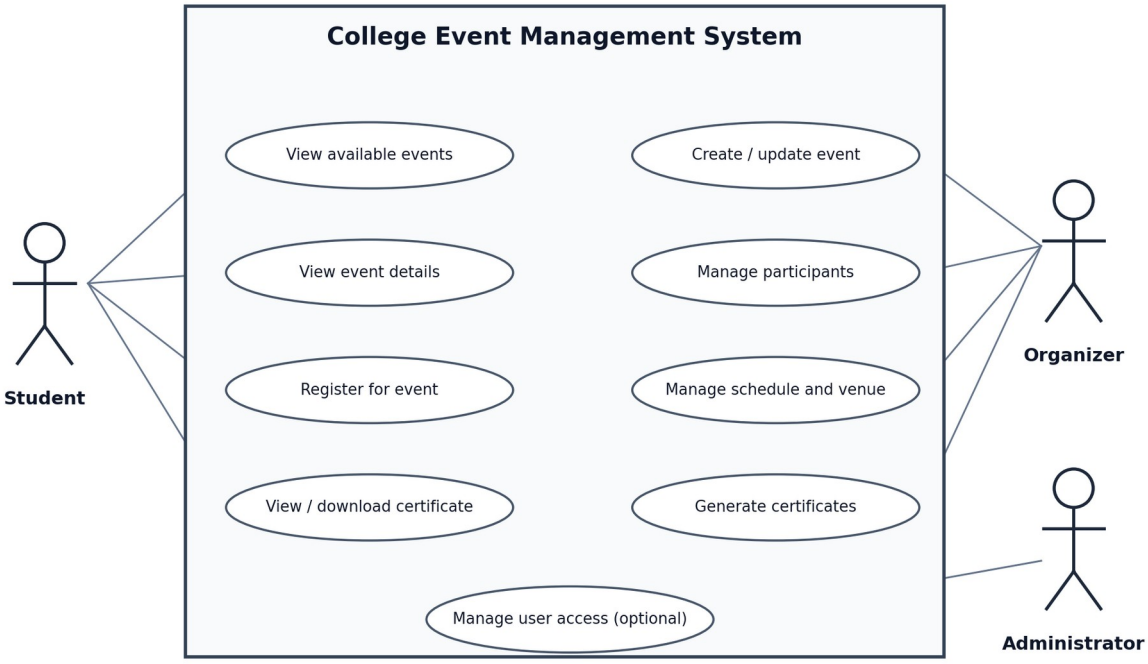
6. Non-Functional Requirements

Non-functional requirements describe the quality and operating expectations of the system rather than individual features. For this academic prototype, the focus is on basic usability, access control, reasonable response, reliable data handling, browser support and ease of maintenance. These are general targets to guide design and testing, not measured performance claims.

ID / Category	Requirement
NFR-01 – Usability	The interface should be simple and understandable. Students should be able to find event information and registration options without confusion, while organizers should be able to manage event details easily.
NFR-02 – Security	Only users with the appropriate access should be able to use organizer or administrator functions. Student-facing functions should not expose restricted management options.
NFR-03 – Performance	Common pages, such as the event list and event details, should load within a reasonable time during normal project use. The project does not require a formal load-testing target at this stage.
NFR-04 – Reliability	Registration and event details should be saved correctly. When an operation cannot be completed, the system should display a clear message instead of silently failing.
NFR-05 – Compatibility	The web application should work in commonly used modern browsers so students and organizers can access it from standard computers or phones using a browser.
NFR-06 – Maintainability	The application should be arranged in a clear way so that event, registration, schedule and certificate functions can be corrected or extended without rewriting unrelated parts.

7. Use Case Modelling

A use case describes a goal that an actor achieves by interacting with the system. The main actors are Student and Event Organizer; Administrator is shown as an optional actor if basic user-access management is included. The diagram gives a high-level view of which functions each actor is expected to use. It is a conceptual model and can be adjusted if the team finalizes a different role structure.



Conceptual diagram; confirm roles and functions with the project team.

The student-side activities mainly cover discovering events, registering and accessing certificates. The organizer-side activities cover creating and maintaining events, managing participants, schedules, venues and certificates. The optional administrator activity is limited to managing user access.

7.1 Main Use Cases

Use case ID	Use case	Actor
UC-01	View events and event details – View the event list and open an event to read its description, date, time and venue.	Student
UC-02	Register for an event – Submit a registration for an event that is accepting participants.	Student
UC-03	View registration status – Check whether the student's registration has been recorded for the selected event.	Student
UC-04	Create or update an event – Create a new event or update existing event information.	Organizer
UC-05	View/manage participant list – Open an event's participant list to review the students who have registered.	Organizer
UC-06	Manage event schedule and venue – Add or update the planned event date, time and venue information.	Organizer
UC-07	Generate participation certificates – Prepare or issue certificates for participants confirmed as eligible.	Organizer

Use case ID	Use case	Actor
UC-08	View/download certificate – Access an issued participation certificate for viewing or downloading.	Student
UC-09	Manage user access (optional) – Manage user accounts or roles, only if this optional function is included.	Administrator

7.2 Example Detailed Use Case – Register for an Event

Actor: Student

Precondition: The student has opened the system. If login is included in the prototype, the student must be signed in before registering.

Main flow:

1. The student opens the list of available events.
2. The student selects an event to view its details.
3. The system displays the event description, date, time and venue.
4. The student selects the Register option.
5. The system checks whether registration is open and whether the student is already registered.
6. If the checks are successful, the system saves the registration and displays a confirmation.

Alternative flow: If registration has closed, the system informs the student and does not save a registration. If the student is already registered for the same event, the system displays a message and prevents a duplicate entry.

Postcondition: A successful registration is saved once and can be shown in the student's registration status. If registration cannot be completed, the student receives a clear reason.

8. User Stories and Acceptance Criteria

A user story expresses a requirement in terms of the person who needs the feature and the benefit it provides. Acceptance criteria are simple checks that help the team decide whether the feature has been implemented correctly. The following stories cover the main student and organizer activities in the proposed system.

US-01 – Browse events

As a student, I want to view available college events so that I can find events I am interested in.

Acceptance criteria: The student can open the event list; each available event shows its title, date and basic details; selecting an event displays its full information.

US-02 – Register

As a student, I want to register for an event so that I can participate.

Acceptance criteria: The student can submit a registration while the event is open; the system records the registration and displays confirmation; a repeated registration for the same event is prevented.

US-03 – Manage event

As an organizer, I want to create and update event details so that students see the correct information.

Acceptance criteria: The organizer can enter the basic event information and save it; the event appears in the listing; changes made by the organizer are reflected in the displayed details.

US-04 – Manage participants

As an organizer, I want to view registered participants so that I can manage event participation.

Acceptance criteria: The organizer can select an event and view the list of students registered for it; the list is associated with the correct event.

US-05 – Manage schedule and venue

As an organizer, I want to maintain event timings and venue details so that participants know where and when the event takes place.

Acceptance criteria: The organizer can save or update the event schedule and venue; the latest saved date, time and venue are displayed with the event.

US-06 – Certificates

As an eligible participant, I want to access my participation certificate so that I can keep proof of participation.

Acceptance criteria: An authorized organizer can issue a certificate to an eligible participant; the participant can access the issued certificate for viewing or downloading.

9. Assumptions and Constraints

- The first version is a basic college project prototype intended to demonstrate the required workflow; it is not intended to be a production-ready platform.
- Students and event organizers are the primary users. Administrator functions may be limited or omitted if they are not needed for the prototype.
- The event organizer will confirm certificate eligibility. Automatic attendance checking or QR-based verification is not assumed in the initial scope.
- Online payments, QR attendance, external SMS integration and dedicated mobile applications are excluded so that the team can focus on the core event-management functions.
- The technology stack, detailed architecture and database structure are design decisions to be finalized by Person 2 and Person 3 in coordination with the team.

10. Requirement Review and Handover

After preparing the initial requirements, Person 1 shares this document with the team for review. Person 2 uses the agreed functions and user roles as input for the UI prototype, system architecture and ER diagram. Person 3 uses the functional requirements and acceptance criteria as a guide while implementing the event, registration and certificate modules and carrying out basic tests. Person 4 uses the agreed scope and major features while preparing the project plan, schedule, risk plan and final documentation.

Before handover is treated as complete, the team should check that the requirements match the assigned PS 2 statement, that each major function has a clear user or purpose, and that no feature has been added outside the agreed CCA scope. Any correction or decision should be recorded in the relevant Jira task or comment. Once the team agrees on the initial requirement set, this document can serve as the shared baseline for design and development. If a requirement changes later, the team should update the document and Jira rather than relying on informal messages alone.

11. Jira Task Mapping

Jira key	Task	Document section
SGC-16	Project Scope Definition	Sections 1–2
SGC-17	Stakeholder Identification	Section 3
SGC-18	Requirement Elicitation	Section 4
SGC-19	Functional Requirements	Section 5
SGC-20	Non-Functional Requirements	Section 6
SGC-21	Use Case Modelling and User Stories	Sections 7–8
SGC-22	Requirements Validation and Handover	Sections 9–10

Note: This is a simple initial draft prepared from the assigned PS 2 problem statement. It is intended to be reviewed and adjusted by the four-member team before final submission. Requirements marked as assumptions should be confirmed.